EBU5376 Digital Signal Processing

LAB 2

Before you start, please read the following information carefully:

General Instructions:

- The .zip file you downloaded from QMPlus should contain the following documents:
  - This lab sheet (needs to be completed)
  - A MATLAB data file that you need to complete the tasks
- Type your answers in the textboxes provided. Figures need to be copied into the textboxes. If you need more space, enlarge the textbox or add a new box.
- Answers outside the textboxes will NOT be marked.
- If you require any technical assistance, please attend the lab session and ask the on-site TA for help.

How to submit:

- Submit one .zip file to QMPlus. The .zip file should contain:
  1. This LAB SHEET converted to a PDF document
  2. The MATLAB script containing all the work you did

Additional grading criteria:

- Your answers in this sheet must include sufficient explanations/derivations. Marks will be deducted if no explanation is given to support the answer.
- Snapshots of AI-generated results will be ignored.
- Your MATLAB script will be checked from the following three aspects:
  - Whether it can **REPRODUCE ALL THE FIGURES** in this sheet
  - Whether you have added appropriate **COMMENTS** to explain the codes
  - Plagiarism
- If the MATLAB script is not submitted, your grade will be capped to 60%.
- Email submission NOT accepted.
- Do NOT plagiarize. Both the MATLAB script and this sheet will be checked for plagiarism.

Question 1 – Filter Response Analysis **[50 marks]**

In this question, you will practice how to understand filter coefficients by visualising and comparing filter responses.

In Lab 1 materials, you are given a MATLAB data file that consists of the filter coefficients of FOUR different filters. For example, the coefficient arrays of **Filter 1** are named $\boldsymbol{a1}$ (output coefficients) and $\boldsymbol{b1}$ (input coefficients), which corresponds to the transfer function coefficients in

$$H(z) = \frac{\sum_{k=0}^{M-1} b_k z^{-k}}{\sum_{k=0}^{N-1} a_k z^{-k}}$$

Similarly, the coefficients of **Filter 2,3,4** are named $\boldsymbol{a2}, \boldsymbol{a3}, \boldsymbol{a4}$ (output coefficients) and $\boldsymbol{b2}, \boldsymbol{b3}, \boldsymbol{b4}$ (input coefficients).

Task 1: Among the four filters, identify the Finite Impulse Response (FIR) filters and the Infinite Impulse Response (IIR) filters. Explain your answers.

[5 marks]

[Type your answers here]

Task 2: Plot the step response of all filters on the same figure. Briefly compare the filters based on the plot.

[10 marks]

[Paste the figures here]

Task 3: Plot the amplitude response (linear scale) of all filters on the same figure. Briefly compare the filters based on the plot.

[14 marks]

[Paste your figure and type your answers here]

Task 4: Plot the amplitude response (decibel scale) of all filters on the same figure. Briefly compare the filters based on the plot.

[7 marks]

[Paste your figure and type your answers here]

Task 5: Estimate the normalised cut-off frequency (in radians) of each filter. Clearly explain how the estimations were obtained for each filter. You may want to include figures or code descriptions to support your explanation when relevant.

[6 marks]

[Type your answers here]

Task 6: Based on the results obtained in the previous tasks, summarize the characteristics of each filter and discuss the preferred use case for each filter by comparing the filter characteristics.

[8 marks]

[Type your answers here]